Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

Homework 2
SF2524

Aleksi Löf & Anton Blomgren
KTH Royal Institute of Technology

November 24, 2025

Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

Matrix Computations for Large-scale Systems
SF2524

Contents

AI Usage Specification **1**

Problem 1 **2**
a) . . . 2
b) . . . 2
c) . . . 3
d) . . . 3

Problem 2 **4**
a) . . . 4
b) . . . 4
c) . . . 4
d) . . . 4

Problem 3 **5**
a) . . . 5
b) . . . 5
c) . . . 5
d) . . . 5
e) . . . 5
f) . . . 5
g) . . . 6

Problem 4 **7**

Problem 5 **8**
a) . . . 8
b) . . . 8

Problem 6 **9**
a) & c) . . . 9
b) . . . 9
d) . . . 9
e) . . . 10

Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

Matrix Computations for Large-scale Systems
SF2524

AI Usage Specification

Level 2:
In this report, AI was used to check for spelling and grammar errors in text. It was also used to localize and find errors in code. It was **NOT** used to generate code or text.

Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

Problem 1

a)

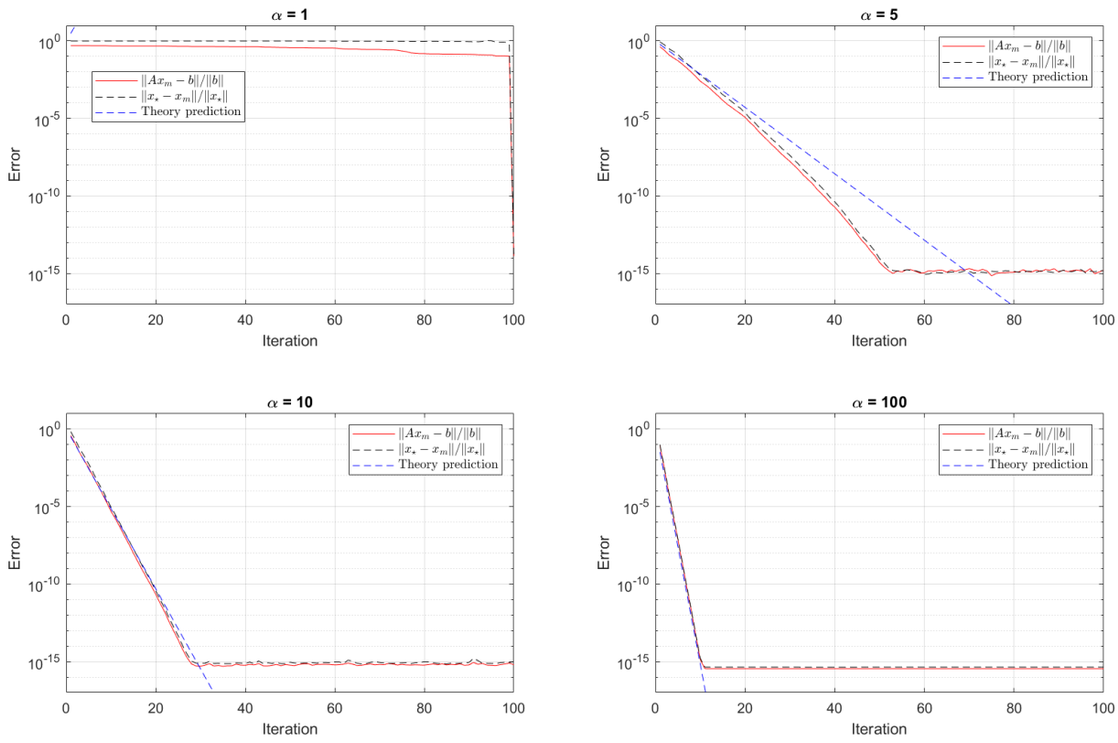


Figure 1: Error for $\alpha = 1, 5, 10, 100$

b)

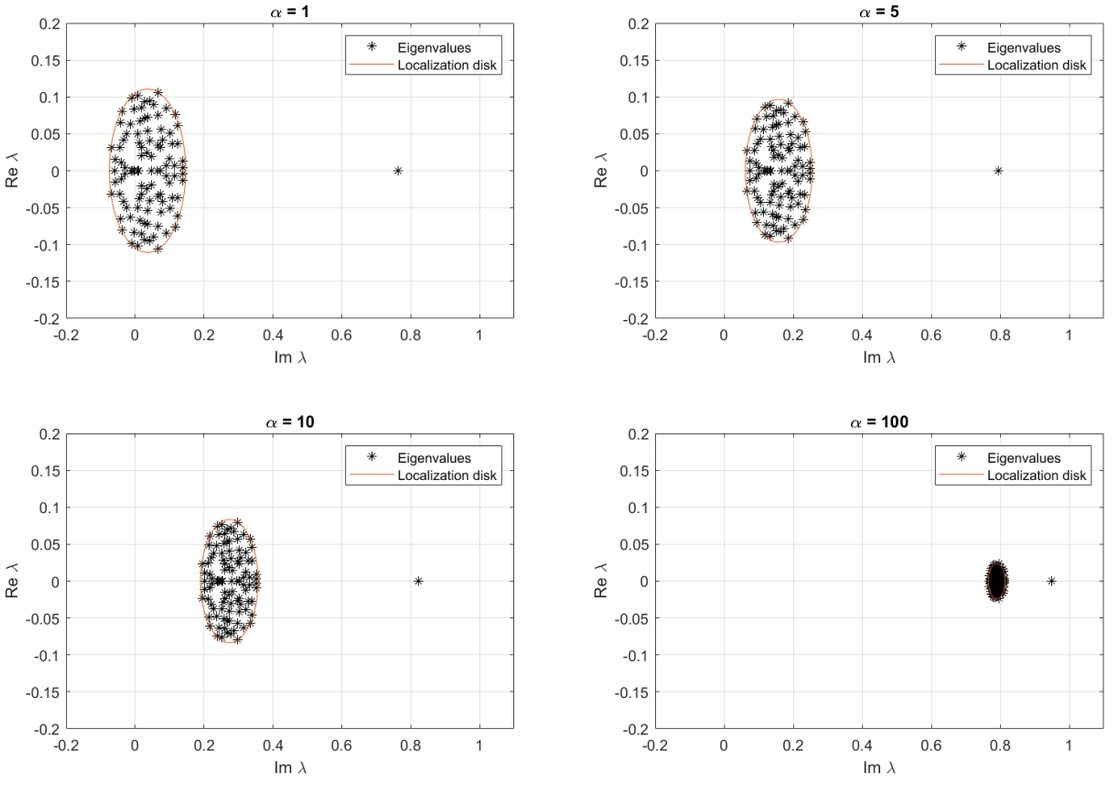


Figure 2: Eigenvalues for $\alpha = 1, 5, 10, 100$

Moving the eigenvalues with α from zero, and creating a tighter cluser, makes the convergence rate $\frac{r}{|c|}$ smaller in value which gives faster convergence.

Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

c)

Tables for $\alpha = 1$

Table 1: GMRES $\alpha = 1$

	$n = 200$		$n = 500$		$n = 1000$	
	resnorm	time	resnorm	time	resnorm	time
$m = 5$	3.91	$6.3 \cdot 10^{-4}$	6.31	$6.15 \cdot 10^{-4}$	9.01	$1.3 \cdot 10^{-3}$
$m = 10$	3.86	$1.7 \cdot 10^{-4}$	6.28	$8.31 \cdot 10^{-4}$	8.94	$2.7 \cdot 10^{-3}$
$m = 20$	3.77	$7.0 \cdot 10^{-4}$	6.23	$1.7 \cdot 10^{-3}$	8.91	$4.6 \cdot 10^{-3}$
$m = 50$	3.54	$2.0 \cdot 10^{-3}$	6.03	$5.1 \cdot 10^{-3}$	8.79	$1.44 \cdot 10^{-2}$
$m = 100$	2.96	$8.7 \cdot 10^{-3}$	5.76	$1.31 \cdot 10^{-2}$	8.56	$3.51 \cdot 10^{-2}$

Table 2: Backslash $\alpha = 1$

	$n = 200$		$n = 500$		$n = 1000$	
	resnorm	time	resnorm	time	resnorm	time
	$2.51 \cdot 10^{-14}$	$3.2 \cdot 10^{-3}$	$6.61 \cdot 10^{-14}$	$1.46 \cdot 10^{-2}$	$1.08 \cdot 10^{-13}$	$7.9 \cdot 10^{-2}$

Tables for $\alpha = 100$

Table 3: GMRES $\alpha = 100$

	$n = 200$		$n = 500$		$n = 1000$	
	resnorm	time	resnorm	time	resnorm	time
$m = 5$	$3.63 \cdot 10^{-6}$	$1.93 \cdot 10^{-4}$	$6.69 \cdot 10^{-5}$	$4.64 \cdot 10^{-4}$	$4.94 \cdot 10^{-4}$	$1.6 \cdot 10^{-3}$
$m = 10$	$5.15 \cdot 10^{-13}$	$2.02 \cdot 10^{-4}$	$8.73 \cdot 10^{-11}$	$7.57 \cdot 10^{-4}$	$4.13 \cdot 10^{-9}$	$2.5 \cdot 10^{-3}$
$m = 20$	$5.26 \cdot 10^{-15}$	$4.86 \cdot 10^{-4}$	$8.94 \cdot 10^{-15}$	$1.8 \cdot 10^{-3}$	$2.09 \cdot 10^{-14}$	$6.6 \cdot 10^{-3}$
$m = 50$	$5.10 \cdot 10^{-15}$	$1.7 \cdot 10^{-3}$	$9.06 \cdot 10^{-15}$	$6.1 \cdot 10^{-3}$	$1.85 \cdot 10^{-14}$	$1.67 \cdot 10^{-2}$
$m = 100$	$5.09 \cdot 10^{-15}$	$5.5 \cdot 10^{-3}$	$9.11 \cdot 10^{-15}$	$1.5 \cdot 10^{-2}$	$1.86 \cdot 10^{-14}$	$4.23 \cdot 10^{-2}$

Table 4: Backslash $\alpha = 100$

	$n = 200$		$n = 500$		$n = 1000$	
	resnorm	time	resnorm	time	resnorm	time
	$2.79 \cdot 10^{-15}$	$2.0 \cdot 10^{-3}$	$7.13 \cdot 10^{-15}$	$1.74 \cdot 10^{-2}$	$1.41 \cdot 10^{-14}$	$7.87 \cdot 10^{-2}$

d)

Observing the row corresponding to $m = 5$ in Table 3, compared with Table 4, one can conclude that if 10^{-5} is sufficient accuracy, GMRES is approximately at least one order of magnitude faster than the Backslash and is therefore ideal to use in this scenario.

Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

Problem 2

a)

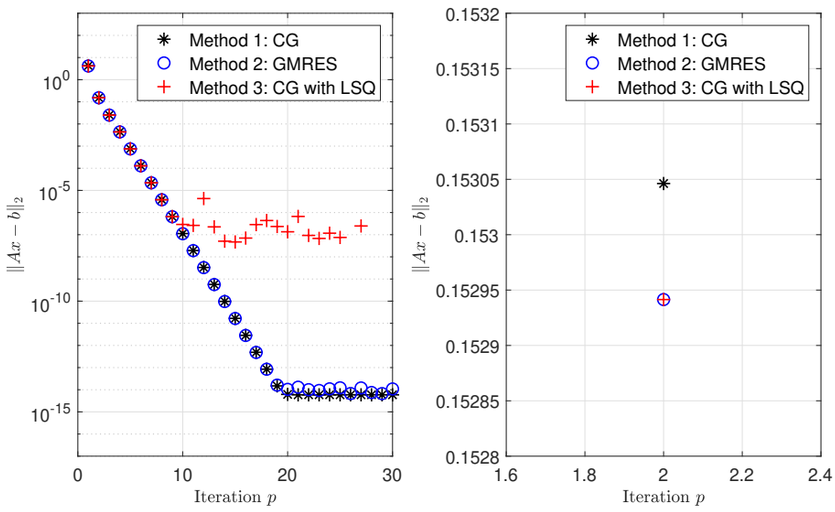


Figure 3: Residual norm depending on iteration for three different methods.

b)

In theory, with the matrix A being symmetric positive definite, GMRES and CG with LSQ should be identical methods with respect to convergence over iterations since we are minimizing the same residual 2-norm over $\mathcal{K}_m(A, b)$. From *Section 2.1.1* in *Block 2*, for GMRES we are constructing an orthogonal matrix Q with column vectors in $\mathcal{K}_m(A, b)$ and use a linear combination of these for the minimizer. For CG with LSQ, to find the minimizer, we use the normal equations to find a linear combination of the CG iterates according to *0.3* in *Background* where the iterates span the same Krylov subspace as GMRES but where the iterates are not orthonormal. In exact arithmetic, the two methods should give the same minimizer but in practice they should differ in iteration.

The reason why CG is not identical to the other two methods is because we are measuring the residual norm with respect to the A^{-1}-norm. This norm is defined if A is symmetric positive definite which it in this case also is. We are therefore with CG minimizing another type of residual norm with respect to A^{-1}, which can also be proven to be equivalent to minimizing the norm of the error with respect to A. With this method we will, in exact arithmetics, not get the same minimizer as the other two methods.

c)

In practice, we will see that GMRES and CG with LSQ behave differently in convergence with respect to iteration. For the right plot in Figure 3 we can see that the lowest residual norm for iteration $p = 2$ is given by the GMRES method and the CG method with LSQ, and we can also notice that they have the same value. At this iteration, when $p = 2$, we are minimizing the residual norm over the Krylov subspace $\mathcal{K}_2(A, b)$ with GMRES. At this stage, the iterates in X can be found in $\mathcal{K}_2(A, b)$ as well, and the LSQ problem is well-posed. We can later see in the left plot in Figure 3, that after $p = 2$ the iterates lose orthogonality and eventually the method loses numerical stability after around $p = 9$ iterations. What we see is a consequence of the use of normal equations when we solve the LSQ problem, where the condition number for the system is now squared. We also have that the iterates from CG become more dependent as we increase p, giving smaller singular values which are then squared due to the usage of normal equations and hence leading to a larger condition number. The CG with LSQ method therefore diverges due to the accumulation of rounding errors and loss of independence in X.

We can also see that CG and GMRES converge similarly but with minor differences in value.

d)

From *Block 2, chapter 2.2* we find that the CG method has complexity $\mathcal{O}(nk)$ while GMRES has $\mathcal{O}(nk^2)$. Comparing these two, CG is more suitable for large scale problems since the computation time per iteration is constant and does not grow to be problematic.

CG uses far less memory than both GMRES and CG + LSQ since it is a short-term recurrence method.

Because the matrix A is symmetric positive definite and considering what has been stated above, the CG method is the best for this problem if A is larger.

Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

Problem 3

a) 201 CPU-socket application and iterative vs direct methods

One example of a linear system is the heat distribution on a CPU socket. This example and many other can be expressed in the form $Ax = b$. In this case it is a FEM problem with a large and sparse matrix A.

When solving $Ax = b$ we can use **direct** or **iterative** methods. Direct methods have a fixed number of operations solving the whole problem with one calculation, while iterative methods give a gradually improving approximation. When we have problems involving large and sparse matrices, they can be both faster and cheaper to solve with iterative methods.

b) 215 GMRES convergence single outlier

In this video the convergence of GMRES is discussed in the setting where all but one eigenvalue is contained in a disk D. Say all eigenvalues but λ_1 are contained in a disk of radius r and with center c, then the convergence can be bounded by:

$$\min_{p \in p_m^0} \max_j \mid p(\lambda_j) \mid \leq \frac{\mid \lambda_1 - c \mid + r}{\mid \lambda_1 \mid} \cdot \frac{r^{m-1}}{\mid c \mid^{m-1}} \tag{1}$$

Where the LHS means that we want to minimize over our constructed polynomials, over the largest eigenvalues. The rate of convergence is the same regardless if we look at the case with an outlier or without, except for a delay of one iteration, which explains the $m - 1$ exponent.

c) 216 Preconditioning for GMRES (part 1)

In many real world scenarios GMRES needs preconditioning to work properly, this will occur when the eigenvalues of the matrix we are investigating are distributed in some unfavorable way. For example a circle around origo in the complex plane.

Then one can tranform the problem by multiplying both sides of the equation, from the left, with a matrix M^{-1}. This is called left preconditioning. If M is designed well, we get faster convergence and cheaper compute. Some common choices of M include a diagonal matrix, upper or lower triangular.

d) 234 Preconditioning CG part 1

When performing left preconditioning on the equation $Ax = b$, we can not simply apply CG, since $B = M_1^{-1} \cdot A$ is not symmetric and positive definite. Also for right preconditioning $AM_2^{-1}M_2x = b$, the B matrix $B = AM_2$ is not generally symmetric and positive definite (SPD). We instead aim to combine these two methods to resolve the issue, our problem is now:

$$M_1^{-1}AM_2^{-1}M_2x = M^{-1}b \tag{2}$$

Where we choose M_1 and M_2 such that $B = M_1^{-1}AM_2^{-1}$ is SPD. We select B and the M matrices such that iB is close to the identity matrix, since that ideal case would mean convergence in one step. It can be shown that if B is to be symmetric, $M_1^{-1} = M_2^{-T}$ needs to hold. If we call $M_2 = R$, B can be expressed as:

$$B = M_2^{-T}AM_2^{-1} = R^{-T}AR^{-1} \tag{3}$$

Then R is selected such that:

$$B = R^{-T}AR^{-1} \approx I \tag{4}$$

e) 235 Preconditioning CG part 2

This video shows how we implement the concepts mentioned in video 234, part d) to the CG algorithm for the so called Preconditioned CG (PCG). We want to apply CG to the system $Ax = b$ by first tranforming it to the form:

$$B\hat{x} = c \tag{5}$$

where $B = R^{-T}AR^{-1}$ and $c = R^{-T}b$. This method gives faster convergence, but does not change the solution space.

f) 250 BiCG method derivation

BiCG is useful since it converges faster than CGN and can solve non-SPD problems unlike standard CG. BiCG is derived from the application of Preconditioned CG to the adjoint linear system

$$\begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} = \begin{bmatrix} \hat{b} \\ b \end{bmatrix} \tag{6}$$

This is a better conditioned problem than that of CGN. The algorithm of BiCG is similiar to that of PCG but we update certain variable in several steps. There is no nice convergence theory for the BiCG method, and the convergence is often erradic.

Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

g) 220 Arnoldi-Tikhonov regularization

If we want to solve $Ax = b$ where b is some unknown vector only given from observations. We can encounter an "ill-conditioned" matrix A, meaning it has a large condition number κ

$$\kappa(A) = \frac{\textbf{maximum singular value of A}}{\textbf{minimum singular value of A}} \tag{7}$$

We then want to approximate x, through the minimal norm solution to $Ax = b$. This can be done through Tikhonov-regularization. With the implementation of the Arnoldi method, giving birth to the Arnoldi-Tikhonov regularization. This is both a smaller minimization problem and also cheaper than the pure Tikhonov regularization.

This method can be used to extract a restored image from an observed noisy one.

Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

Problem 4

From *Theorem 2.2.7* and *Corollary 2.1.5* in *Block 2*, we have, together with $||Ax_m - b||_{A^{-1}} = ||x_m - x_*||_A$ and $q \in P_m^0$, that the bound is given by

$$\frac{||x_m - x_*||_A}{||x_0 - x_*||_A} \leq \min_{p \in P_m^0} \max_{i=1,\dots n} |p(\lambda_i)| \leq \max_{i=1,\dots n} |q(\lambda_i)|. \tag{8}$$

Since we have two outlier eigenvalues $\lambda_1 = 10$ and $\lambda_2 = 10.5$, we want to normalize these to have polynomial value zero, meaning $q(10) = 0$ and $q(10.5) = 0$, similar to what is done in *Corollary 2.1.6* in *Block 2* but with two outlier eigenvalues. Thus we have

$$|q(\lambda_i)| \leq \frac{|\lambda_1 - c| + r}{|\lambda_1|} \frac{|\lambda_2 - c| + r}{|\lambda_2|} \left(\frac{r}{|c|}\right)^{m-2} = C \left(\frac{r}{|c|}\right)^{m-2}, \tag{9}$$

where $c = 2.5$ and $r = 0.5$ in the interval $I = [2, 3]$ and C is independent of m.

From *Theorem 2.2.8* in *Block 2*, we have that since A is symmetric and positive definite, one can use *Chebyshev Polynomials* to derive the following condition number bound with (8) and (9)

$$\frac{||x_m - x_*||_A}{||x_0 - x_*||_A} \leq 2C \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right)^d = \varepsilon, \tag{10}$$

where $d = m - 2$ since C takes care of the two iterations for the outliers. We thus also have $\kappa = \lambda_{max}/\lambda_{min}$ where $\lambda_{max} = 3$ and $\lambda_{min} = 2$ correspond to the largest and smallest eigenvalues in the interval $I = [2, 3]$ respectively. We also want $\varepsilon \leq 10^{-7}$ for the error to be reduced by a factor 10^7.

Rearranging the right side of the inequality in (10) with $d = m - 2$ and taking the logarithm of both sides we get

$$(m-2) \log\left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right) \leq \log\left(\frac{\varepsilon}{2C}\right) \quad \Leftrightarrow \quad m \geq \frac{\log\left(\varepsilon/2C\right)}{\log\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)} + 2. \tag{11}$$

Finally, with specific values we get

$$m \geq \frac{\log\left(10^{-7}/(2 \cdot 0.65)\right)}{\log\left(\frac{\sqrt{1.5}-1}{\sqrt{1.5}+1}\right)} + 2 \approx 9.1. \tag{12}$$

For the iterations we only look at integer steps, meaning if we choose an integer value larger than $m = 9.1$ achieved in (12), we will satisfy the criterion for the error being reduced by a factor 10^7. Thus, it takes at least $m = 10$ iterations with CG to reduce the error by a factor 10^7. $\square$

Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

Problem 5

a)

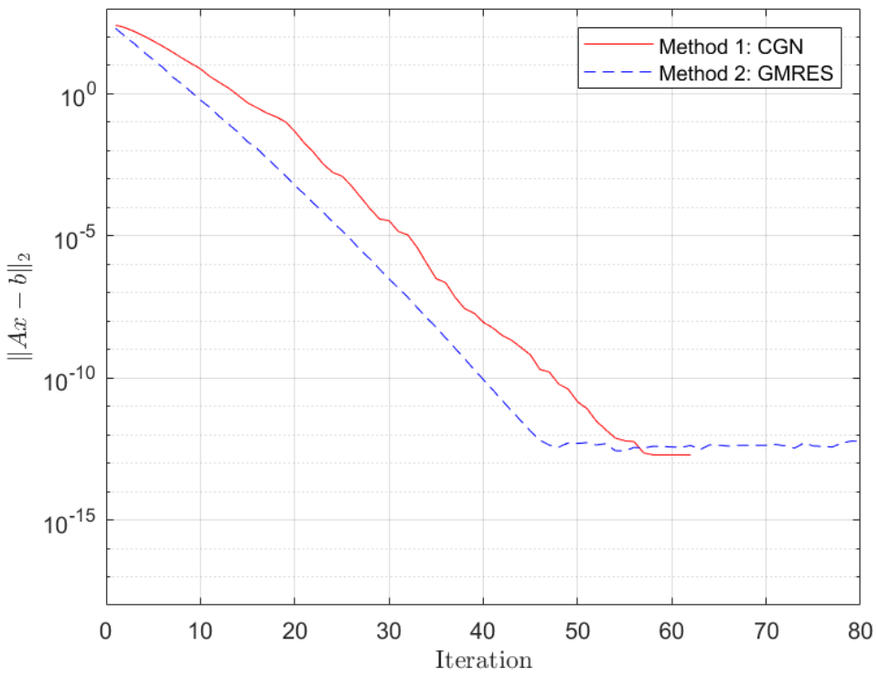


Figure 4

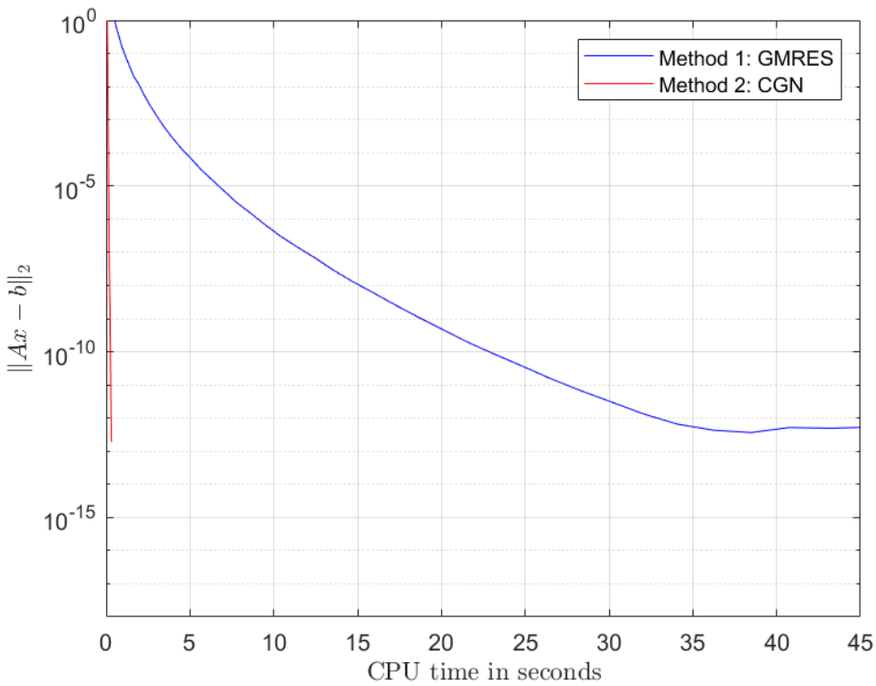


Figure 5

b)

For convergence, CGN performs more iterations than GMRES, but does it significantly faster concerning CPU time. According to *Section 2.3.2* in *Block 2*, GMRES minimizes the residual over the subspace $\mathcal{K}_m(A, b)$ while CGN minimizes the residual over the subspace $\mathcal{K}_m(A^TA, A^Tb)$. The claim is then that the CGN subspace has worse approximation properties. This can explain why CGN needs more iterations and is thus more expensive (it performs two matrix multiplication operations per iteration). Although the fast convergence in CPU time for CGN is a product of the fact it is a short-term recurrence method, meaning it has a constant cost per iteration and a small memory requirement.

Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

Problem 6

a) & c)

The following results were gathered for part a) and c) combined.

Backslash		$Ax = b$	
$X = 1$	$X = 2$	$X = 1$	$X = 2$
0.0687 [s]	0.9356 [s]	$2.07 \cdot 10^{-4}$ [s]	0.001 [s]

b)

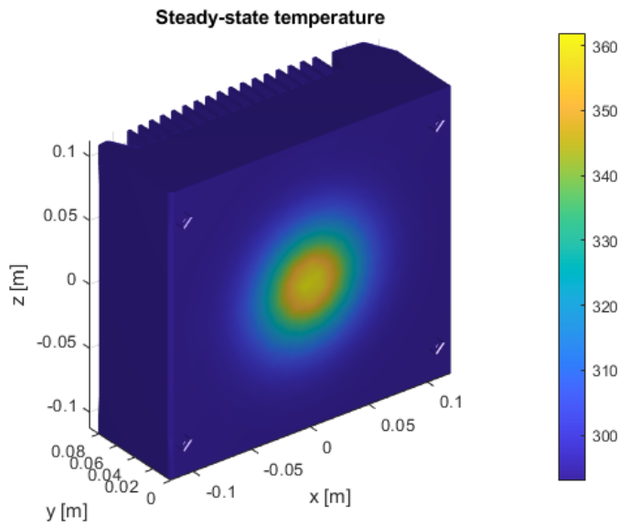


Figure 6: Heatsink

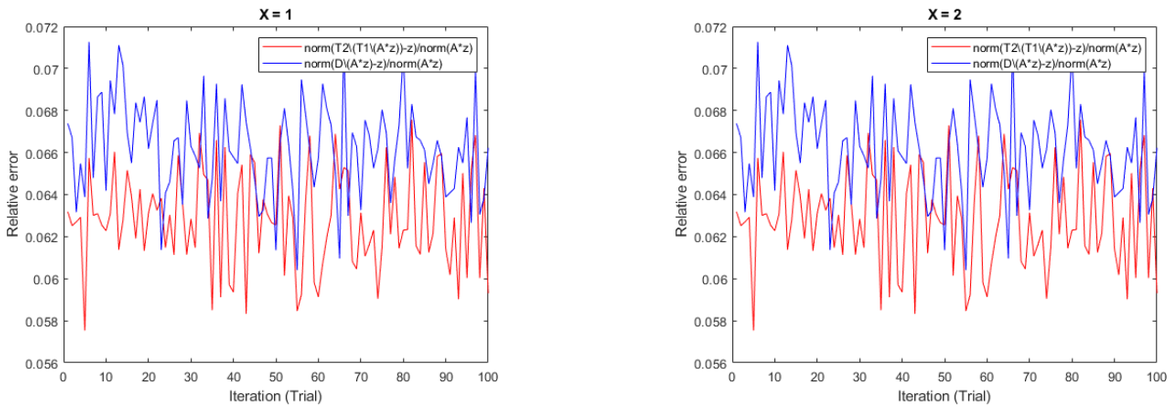
d)

Timetable for d) with average CPU-times:

$A^{-1} \approx D^{-1}$		$M^{-1} = T_2^{-1}T_1^{-1} \approx A^{-1}$	
$X = 1$	$X = 2$	$X = 1$	$X = 2$
$2.08 \cdot 10^{-4}$ [s]	$2.3 \cdot 10^{-3}$ [s]	$5.28 \cdot 10^{-4}$ [s]	$4.6 \cdot 10^{-3}$ [s]

.

In the above table it is observed that the diagonal preconditioner is approximately twice as fast.



(a) (b)

Figure 7: Norms for the two preconditioners.

From figure 7 we conclude that the preconditioner M^{-1} is more accurate since it has a lower overall average norm for both cases.

Aleksi Löf
Anton Blomgren
aleksil@kth.se
ablomg@kth.se

e)

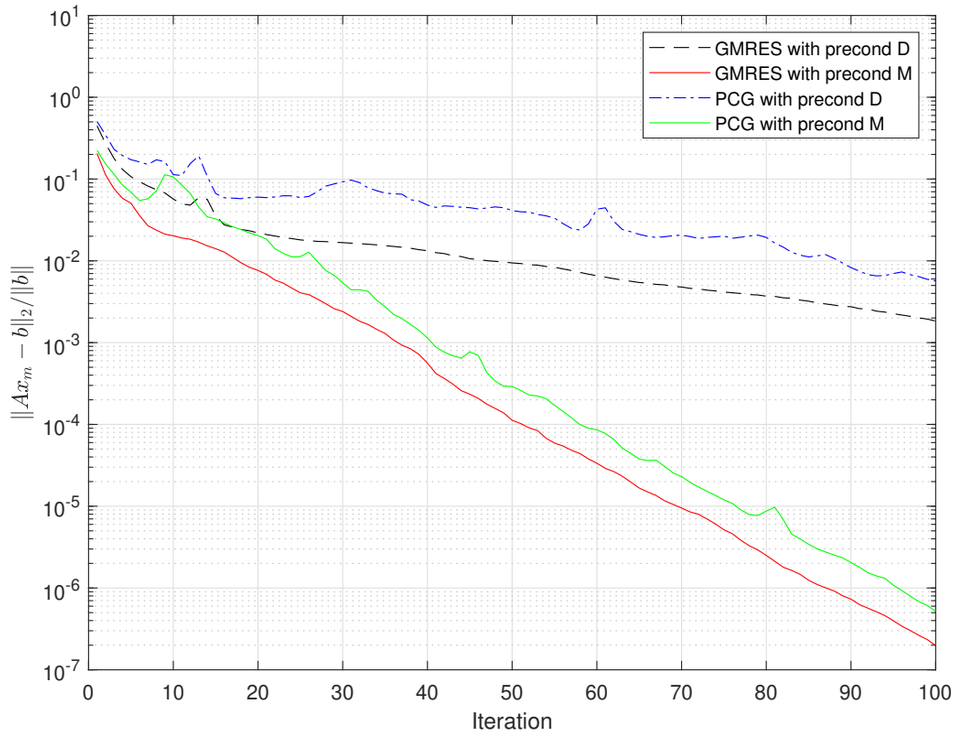


Figure 8: Accuracy over iterations for GMRES and PCG with different preconditioners.

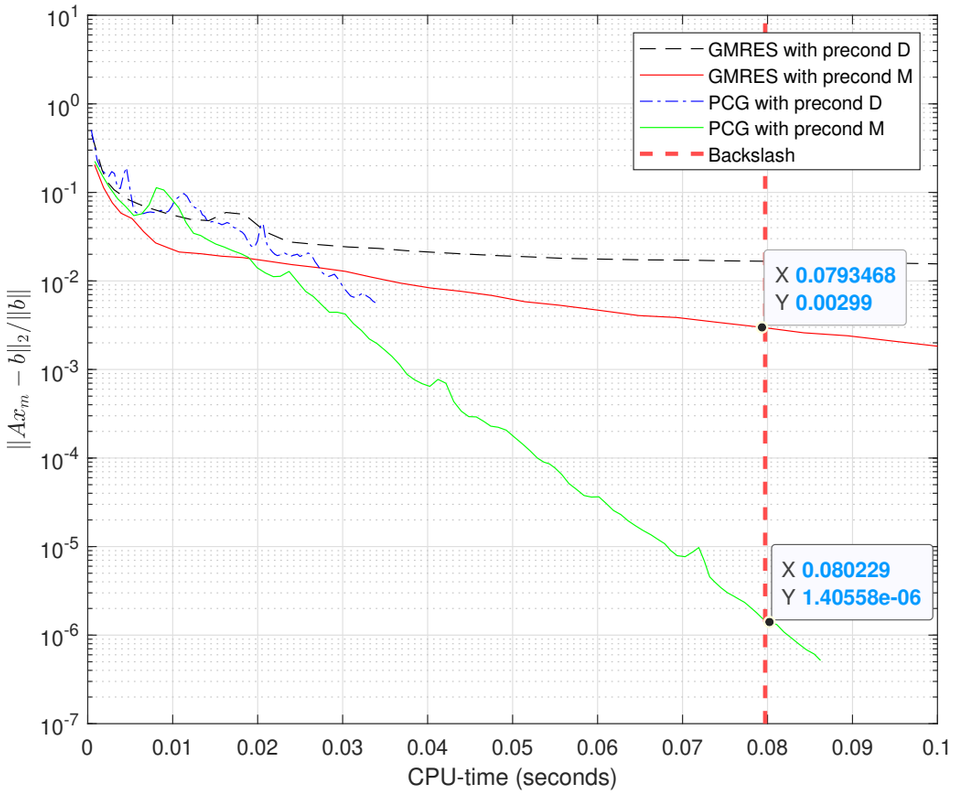


Figure 9: Where Backslash meets other preconditioned methods regarding CPU time for 100 iterations.

In Figure 8 we clearly observe M^{-1} preconditioning reaches better accuracy with both methods. In particular, the GMRES method with M^{-1} as a preconditioner achieves the overall best accuracy after 100 iterations.

In Figure 9 we observe that for the GMRES method with preconditioning M^-1, with the best convergence concerning iteration, the method is a better idea to use than *Backslash* up to an accuracy $\approx 10^{-3}$. Worth to note is that we can also see that the CG method with preconditioning M^-1, which had the second best convergence with respect to iteration, is a better idea to use than *Backslash* up to an even greater accuracy $\approx 10^{-6}$.